Few-Shot Open-Set Hyperspectral Image Classification With Adaptive Threshold Using Self-Supervised Multitask Learning

Caihong Mu, *Senior Member, IEEE*, Yu Liu, Xiangrong Yan, Aamir Ali, and Yi Liu

***Abstract*— Existing hyperspectral image (HSI) classification methods rarely consider open-set classification (OSC). Although some reconstruction-based methods can deal with OSC, they lack adaptive threshold strategies and heavily rely on the labeled samples. Therefore, this article proposes a self-supervised multitask learning (SSMTL) framework for few-shot open-set HSI classification, including three stages: pretraining stage (PTS), fine-tuning stage, and testing stage. The model consists of three modules: data diversification module (DDM), 3-D multiscale attention module (3D-MAM), and adaptive threshold module (ATM), as well as a backbone network: dense feature pyramid network (DFPN). In the PTS, we construct a self-supervised reconstruction task with unlabeled samples for model initialization, where DDM aims to improve the robustness of the model and 3D-MAM applies 3-D multiscale convolution to focus on key information spatially and spectrally. In the fine-tuning stage, we further optimize the model with a few labeled samples based on both reconstruction task and classification task, where ATM implements adaptive threshold strategies based on uncertainties of predicted probability and reconstruction loss, and DFPN is helpful to retain the detailed information. The experimental results on three common HSI datasets show SSMTL performs significantly well and even surpasses many advanced closed-set and open-set HSI classification methods.**

***Index Terms*— Adaptive threshold, hyperspectral image (HSI), multitask learning, open-set classification (OSC), self-supervised learning.**

I. Introduction

With the rapid development of remote sensing platforms and sensors, remote sensing technology has gradually become a scientific field of concern. Among the commonly used remote sensing images, hyperspectral images (HSIs) have many wide application scenarios in agriculture, the military, meteorology, and other fields due to the rich spatial and spectral information contained in HSIs [1], [2]. HSI classification is a remote sensing image processing technology that classifies ground objects pixel by pixel, which can assign an independent ground object label to each pixel in the HSIs, such as grassland, road, forest, soil, building, and river. In fact, HSI classification is a fundamental and important image processing technique, which is widely applied in the scenarios involving HSIs [3], [4], [5].

Most of the early HSI classification methods focused on rich spectral features of HSIs, and many traditional machine learning-based methods were proposed in the past decade, such as the support vector machine (SVM) [6], [7], k-nearest neighbor (KNN) [8], the multinomial logistic regression [9], and the Bayesian estimation method [10], which brought a lot of inspiration and experience to the later research. At the same time, since the effect of the classification model is easily affected by the redundant information in the HSIs, some researchers focus on dimensionality reduction of the high-dimensional spectral information of HSIs to obtain important features for classification. Some classic examples of such methods are the principal component analysis (PCA) [11], [12], the linear discriminant analysis (LDA) [13], [14], [15], and sparse preserving projection (SPP) [16], enabling the classification model to obtain a more refined feature representation. Due to the limitations of natural environment such as illumination and atmosphere, the quality of HSIs obtained in the real world is not always high, which poses a severe challenge to the learning ability of the model. In recent years, with the rapid development of deep learning, deep neural networks (DNNs) have been widely used in the field of computer vision. Massive HSIs were used to train various DNNs, such as convolutional neural networks (CNNs) [17], [18], [19], [20], [21], [22], [23], [24], graph convolutional networks (GCNs) [25], [26], [27], recurrent neural networks (RNNs) [28], [29] and transformer [30], [31], [32], [33] for HSI classification. Benefiting from the powerful feature extraction ability of DNNs, the deep learning-based methods have become increasingly prevalent for HSI classification. In addition, since the features extracted by DNNs are too complex, attention mechanism (AM) modules have been applied to focus on key information and weaken redundant information, effectively reducing the burden of the network and enhancing the feature representation [34], [35]. To further improve the performance of the model, the organic combination of different neural networks also produced a number of new network structures with high performance for HSI classification [36], [37], [38]. Most of these methods are

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supervised, which usually use a large amount of labeled data to train the model. Considering the high cost of acquiring labeled samples, more and more researchers gradually pay attention to few-shot HSI classification and propose many different learning paradigms, such as semi-supervised learning [39], unsupervised learning [40], [41], self-supervised learning [42], [43], [44], transfer learning [45], [46], meta-learning [47], [48], multitask learning [49], and multiview learning [50], [51] for few-shot HSI classification.

These above methods generally assume that the categories of ground object used for testing are all included during the training process (i.e., closed-set environment). In the real world, it is common that a novel category of unknown ground objects will appear during the testing process (i.e., open-set environment). Unfortunately, the existing open-set or few-shot open-set HSI classification methods are still very limited and have not attracted wide attention. A simple way to transform a closed-set model into open-set one is to set a confidence thresholding directly in the closed-set model, where pixels with lower classification confidence are considered to belong to novel class. However, the effect of this method depends heavily on the setting of confidence thresholding, and it cannot achieve ideal results due to the limitation of hard threshold. Besides, in some cases, lower classification confidence does not necessarily mean a novel class. To further improve the classification accuracy (CA), a multitask deep learning method for open world (MDL4OW) was proposed based on reconstruction strategy [52]. Considering the semantic feature extracted from the shallow networks, a spectral–spatial latent reconstruction (SSLR) learning was designed to further make use of the spectral–spatial information based on the framework of MDL4OW [53]. Nonetheless, the above two methods [52] and [53] need to manually set the threshold value for both the tail data and the cumulative distribution function of the generalized Pareto distribution. The empirical threshold of parameters cannot automatically adapt to the changes in different data and requires a lot of time to adjust. Moreover, the output features of the high-level layers of DNNs may lose the details of novel class, resulting in the failure of the model to detect novel samples [54], [55]. Meanwhile, since the label information used in the training process belongs to the known classes, the samples of novel class will tend to be classified as the known classes when DNNs perform backpropagation with too many training epochs.

Recently, some researchers also used reciprocal points learning to model extra-category space, which aimed to reduce the risk of open space [56]. The trained model could increase the distance between different classes and distinguish novel class from known classes efficiently. However, compared with the closed-set methods, the standard deviations of classification accuracies obtained by the existing open-set methods were generally larger, indicating that the existing method could not guarantee the robustness of the model. Moreover, the above methods are still stuck in the scenario of at least 20 or 30 labeled samples per class. Nevertheless, many HSI classification tasks often have fewer labeled samples, with only 5–10 labeled samples per class available for training [57], [58].

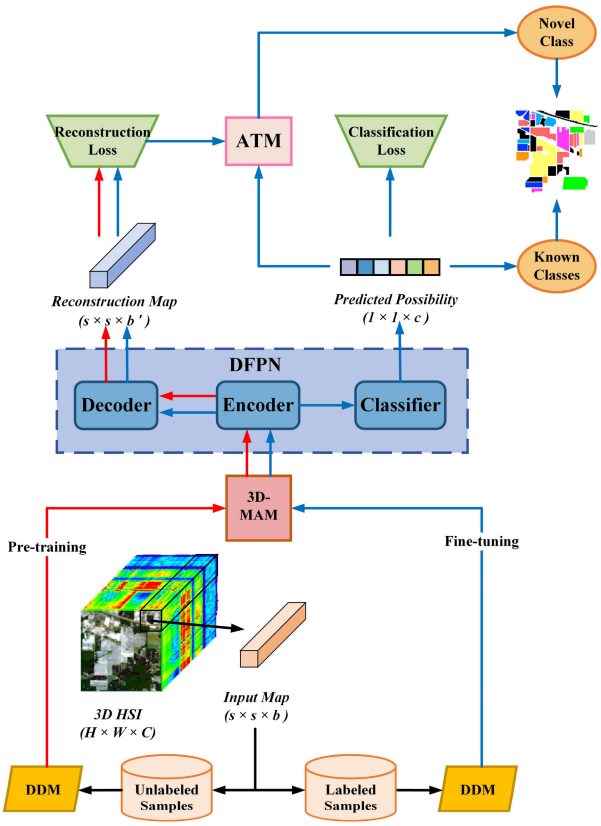


Fig. 1. Framework of the proposed SSMTL. The model is trained by the reconstruction task without using the classifier in the PTS, while it will include the classifier in the fine-tuning stage.

To address the above problems, we propose a self-supervised multitask learning (SSMTL) framework for few-shot open-set HSI classification. As shown in Fig. 1, SSMTL mainly involves three stages, three modules, and a backbone network. The three stages include pretraining stage (PTS), fine-tuning stage, and testing stage. The three modules are data diversification module (DDM), 3-D multiscale attention module (3D-MAM), and adaptive threshold module (ATM). The backbone network is named dense feature pyramid network (DFPN). The proposed framework SSMTL is based on multitask learning including classification task and reconstruction task, where reconstructing deep features to represent shallow features as the auxiliary task is a common self-supervised learning method from unlabeled data [52].

To be specific, SSMTL is mainly composed of PTS, fine-tuning stage, and testing stage, where the settings of DDM and classifier in three stages are different. In the PTS, DDM performs spectral random sampling (SRS) and data augmentation on all unlabeled samples as input, while in the fine-tuning stage, DDM performs PCA dimensionality reduction and data augmentation on a few labeled samples as input. In the testing stage, the structure is the same as that of fine-tuning stage. SRS is an effective method to improve the robustness of the model at the data level, while it has a certain sampling time cost [59]. At the same time, we design a 3D-MAM to further improve the performance of the model. Compared with 2D-CNN, 3D-CNN can fully combine the spatial and spectral information, and

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the introduction of multiscale convolution can obtain feature maps of different scales, which is beneficial to fully enriching the spatial and spectral features. In addition, we design the DFPN as the backbone network for the feature extraction and feature reconstruction. The dense network can reuse the shallow features of the HSIs for the feature extraction, and the structure of the feature pyramid can provide shallow features to assist the feature reconstruction, which effectively avoids the information loss accompanying the deep convolution neural network. Finally, inspired by the open-set semantic segmentation method of natural images [60], we design ATM based on the difference of uncertainty between known classes and novel class, which can automatically determine the parameter threshold by considering the entropy distribution of both reconstruction loss and predicted probability.

The main contributions of this article are as follows.

1) An efficient SSMTL framework is proposed for few-shot open-set HSI classification. In the PTS, we initialize the parameters of the model by minimizing the reconstruction loss. In the fine-tuning stage, we fine-tune the parameters of the model by minimizing the reconstruction loss and classification loss.
2) A 3D-MAM is proposed to focus on key information and weaken redundant information. With 3-D convolution layer to join the spatial and spectral features of different scales, the 3D-MAM can obtain the weighted multiscale spatial–spectral features, and establish the long-distance dependence relationship of effective information.
3) A DFPN is designed to extract the deep semantic feature while retaining the shallow semantic feature. DFPN helps to balance the relationship between the known classes and the novel class with the limited depth of the network and the training epochs of the model.
4) An ATM is proposed to dynamically generate parameter thresholds. As an entropy-based (or uncertainty-based) method, ATM provides an automatic strategy to determine the parameter threshold to distinguish the novel class from the known classes, which avoids the tedious process of manually adjusting the parameter to determine the threshold.

The rest of the article is arranged as follows. In Section II, the backgrounds of open-set classification (OSC) are further introduced. In Section III, SSMTL is described in detail. In Section IV, the experimental verification and analysis are shown. In Section V, the conclusion of the article is summarized comprehensively and some issues for future improvements are proposed.

II. Backgrounds

As mentioned above, HSI classification technology has made great progress, and advanced network structures and learning paradigms are constantly proposed. However, the vast majority of HSI classification methods assume that the categories of ground object used for testing are all included in the training process, i.e., it is a kind of closed-set environment. The corresponding classification task is called closed-set classification (CSC) [50]. Obviously, this assumption is not in line with the real classification scenarios, because it is impossible or very difficult to manually label categories for all kinds of ground objects in the real world. Open-set environment refers to the scenario where unknown ground objects with novel categories will appear during the testing process, and the corresponding classification task is called OSC.

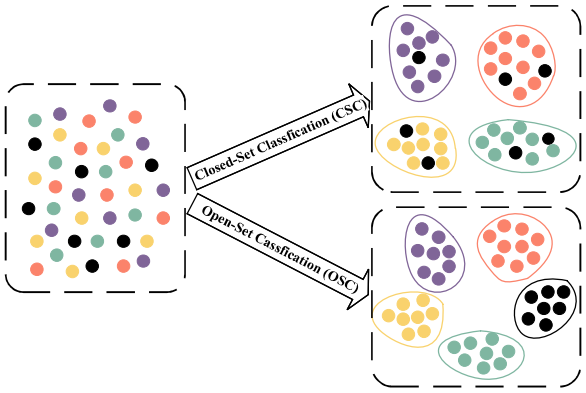


Fig. 2. Representation of OSC and CSC. The black circles denote the novel class, while other colored circles denote the known classes.

For most HSI classification tasks, we only focus on part of the ground object categories with obvious characteristics or application requirements, and too many ground object categories may interfere with the analysis of HSIs. Therefore, in the testing process of OSC, all unknown ground feature categories are usually considered as a novel class. As shown in Fig. 2, the classification performance of closed-set HSI classification methods always degrades in the open-set environment, because the model without prior knowledge of the novel class will mistake the unknown ground objects for some known classes. Therefore, open-set HSI classification has realistic significance for improving CA. In fact, natural images also suffer from OSC, and many related works have been proposed. These methods are mainly based on generative and discriminative methods to find the decision boundary between the novel class and the known classes [61], [62]. Unlike the natural image classification, HSI classification is a pixel-level task with fewer labels and classes, which results in HSIs having lower openness. Therefore, open-set HSI classification often needs to consider the few-shot problem.

As a kind of generative methods, the reconstruction-based approach is effective for few-shot open-set HSI classification, which constructs shallow semantic features from deep ones. And the constructed features of the novel class and the known classes can be distinguished by setting an appropriate criterion, which is a kind of time-consuming task without adaptive threshold strategies. Besides, the existing methods still lack a reasonable network structure to fully learn the HSI features with the shallower neural network and fewer training epochs of the model, which is beneficial to curbing the loss of feature details and the misclassification of novel samples. Furthermore, the robustness and stability of the model deserve further improvement with fewer labeled samples.

III. Method

In this section, we first review some components of SSMTL framework, including DDM and 3D-MAM. Then,

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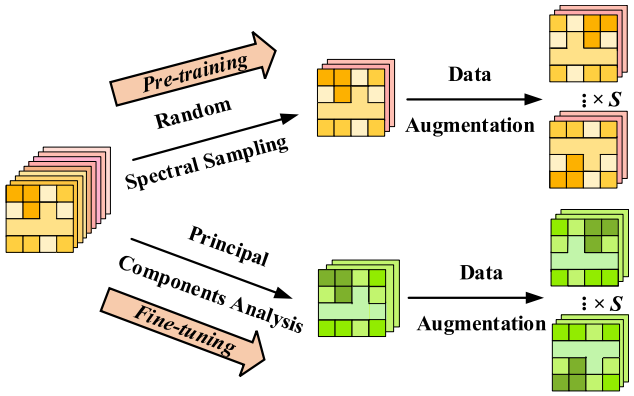


Fig. 3. Schematic of DDM. There are S types for data augmentation in the DDM.

we describe the details of our proposed DFPN and loss function, including reconstruction loss and classification loss. Finally, we introduce our proposed ATM based on the uncertainty of reconstruction loss and predicted probability.

A. Data Diversification Module

Like common HSI classification methods, SSMTL takes HSI patches as input [63]. However, in the PTS, we choose SRS instead of PCA for spectral dimensionality reduction on the patches. SRS has been proven to be effective in improving the robustness of HSI classification models, which produces a randomized effect by randomly sampling along the spectral band pixel by pixel [59]. It should be noted that the SRS is not for the whole image but for the patch centered at each pixel. Let $X \in \mathbb{R}^{s\times s\times b}$ represent the input patches, where $s \times s$ and b denote the patch size and number of bands, respectively. As shown in Fig. 3, after randomly sampling b' spectral bands to generate a group of patches $X_{\text{pretrain}} \in \mathbb{R}^{s\times s\times b'}$ as the input of PTS, data augmentation with S types is performed on the patches of unlabeled samples, such as horizontal, vortical, vertical, and diagonal augmentation. Similarly, after applying PCA to generate a group of patches $X_{\text{tuning}} \in \mathbb{R}^{s\times s\times b'}$ as the input of fine-tuning stage, data augmentation is also performed on the patches of labeled samples.

Why are different settings adopted for the PTS and the fine-tuning stage in DDM? As mentioned above, the unlabeled samples are used as the input for the PTS, where unlabeled data is sufficient and no label information is needed. Therefore, on the premise of sufficient data for pretraining, it is meaningful and acceptable for DDM to increase the diversity of data by SRS that will discard part of spectral data. However, there are only limited labeled data available in the fine-tuning stage. If SRS is also applied in the DDM for fine-tuning, the few training data after discarding part of the spectral data will not support the model to fully learn the label information of HSIs. On the contrary, using PCA for spectral dimensionality reduction without losing limited and important labeled data is clearly a more sensible choice.

B. Three-Dimensional Multiscale Attention Module

Many existing HSI classification methods use the AM to improve the performance of CNN and achieve ideal results [20], [21], [22], [23]. However, multiscale AM is few applied to HSI classification. In fact, the size of the feature maps has an important impact on the judgment of edge features, which is also a common way to improve the accuracy of the model. In the field of natural image processing, efficient pyramid split attention (EPSA) is an effective multiscale channel attention structure, which not only extracts the spatial feature information of different scales but also establishes the long-term dependence of channel attention [64]. Inspired by EPSA, the proposed 3D-MAM also focuses on extracting and combining spatial–spectral information at different scales.

As shown in Fig. 4, 3D-MAM mainly includes the following steps: first, the outputs of DDM are fed into a 3-D convolution layer with different convolution kernels to extract feature maps with different scales, which are concatenated to generate the multiscale feature map. Second, multiscale spatial and spectral attention vectors are obtained by extracting the attention weights of multiscale feature maps through the spatial attention block and spectral attention block, respectively. Next, soft attention is used to adaptively calibrate the multiscale spatial and spectral attention vectors, where the multiscale spatial attention weights and the multiscale spectral attention weights are obtained, respectively. Then, the multiscale spatial and spectral attention weights are used to multiply the corresponding feature map in turn. Finally, the weighted feature map with rich spatial–spectral information is output.

1) Branches of 3-D Multiscale Convolution: Different from natural images, an HSI not only has spatial information but also has rich spectral information. Three-dimensional convolution layer can fully combine the spatial and spectral information of HSI. As shown in Fig. 4, we extract multiscale feature information of 3-D HSI patches through multiple branches, where the size of the convolution kernel is different in each branch. Specifically, 3-D convolution layer compresses the input tensor $X_{\text{input}} = X_{\text{pretrain}}$ or X_{tuning} of size $s \times s \times b'$ both spatially and spectrally to obtain multiscale feature information with different spatial and channel sizes, where the number and size of convolution 3-D kernels is q and $k_i \times k_i \times k_i, i = 1, 2, \ldots, t$, respectively. Eventually, t branches will output tensors of size $q \times s \times s \times b'$. Note that there is padding in the convolution process, so the spatial sizes of feature images are always $s \times s$. Therefore, the multiscale feature generation process can be expressed as follows:

$$\begin{aligned} F_i &= \text{Conv3d}(k_i \times k_i \times k_i)\left(X_{\text{input}}\right), \quad i = 1, 2, \ldots, t \\ F &= \text{Concat}(F_1, F_2, \ldots, F_t) \end{aligned} \tag{1}$$

where $F_i \in \mathbb{R}^{q\times s\times s\times b'}$ represents the feature map on the ith branch, $F \in \mathbb{R}^{q\times s\times s\times (b'\times t)}$ represents the final multiscale feature map, and Concat stands for concatenation operation.

2) Spatial Attention Block and Spectral Attention Block: After obtaining multiscale feature map $F \in \mathbb{R}^{q\times s\times s\times (b'\times t)}$, spatial attention block and spectral attention block are used to extract attention weights, which mainly consist of average pooling layer, max pooling layer, and convolution layer, as shown in Fig. 5. First, the average pooling layer and max pooling layer perform pooling operation on $F_i \in \mathbb{R}^{q\times s\times s\times b'}$ in the spatial and the spectral dimension to generate the spatial attention vector $P_i \in \mathbb{R}^{q\times s\times s\times 1}$ and spectral attention vector

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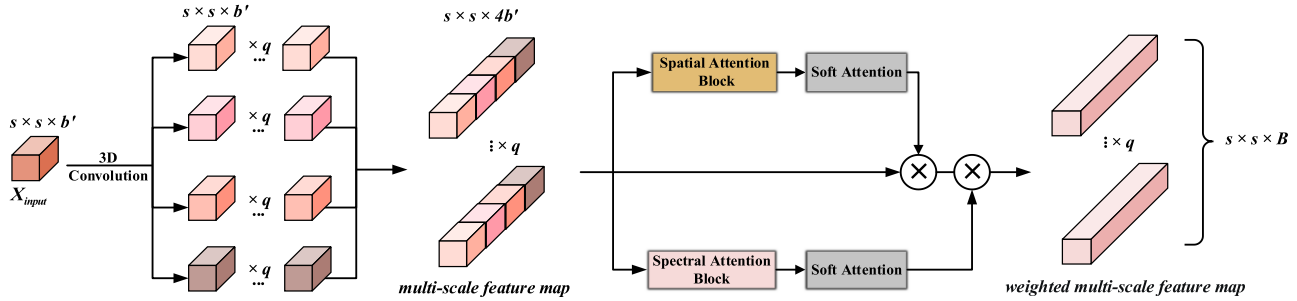


Fig. 4. Schematic of 3D-MAM. The number t of branches here is set to 4, where q denotes the number of 3-D convolution filters on each branch. Finally, the number of spectral channels equals to $B = b' \times t \times q = b' \times 4 \times q$.

$Z_i \in \mathbb{R}^{q\times 1\times 1\times b'}$, respectively. Then, the convolution layer fuses the cross-dimensional vectors without destroying the original multiscale channel, after which t groups of attention vectors corresponding to feature maps of different branches are obtained. It is important to note that the features are not mixed from different scales, and the convolution layers only fuse the context information from a single scale. Mathematically, spatial attention vector and spectral attention vector can be expressed as follows:

$$\begin{aligned} P_i &= \text{Spa-Att}(F_i), \quad i = 1, 2, \ldots, t \\ Z_i &= \text{Spe-Att}(F_i), \quad i = 1, 2, \ldots, t \end{aligned} \tag{2}$$

where Spa-Att and Spe-Att represent the process of obtaining spatial and spectral attention vectors from input feature maps at different scales; $P_i \in \mathbb{R}^{q\times s\times s\times 1}$ and $Z_i \in \mathbb{R}^{q\times 1\times 1\times b'}$ represent the spatial attention vector and spectral attention vector on the ith branch, respectively.

3) Soft Attention: As for the attention weights from different branches with different scales, the soft attention adaptive selection is realized based on Softmax, and the attention weights are calibrated

$$\begin{aligned} \tilde{P}_i &= \text{Softmax}(P_i) = \frac{\exp(P_i)}{\sum_{i=1}^{t}\exp(P_i)} \\ \tilde{Z}_i &= \text{Softmax}(Z_i) = \frac{\exp(Z_i)}{\sum_{i=1}^{t}\exp(Z_i)} \end{aligned} \tag{3}$$

where $\tilde{P}_i \in \mathbb{R}^{q\times s\times s\times 1}$ and $\tilde{Z}_i \in \mathbb{R}^{q\times 1\times 1\times b'}$ represent the spatial attention weight and spectral attention weight of ith branch, respectively. Then, the whole multiscale attention weight vectors are obtained by concatenation operation

$$\begin{aligned} \tilde{P} &= \text{Concat}\left(\tilde{P}_1, \ldots, \tilde{P}_t\right) \\ \tilde{Z} &= \text{Concat}\left(\tilde{Z}_1, \ldots, \tilde{Z}_t\right) \end{aligned} \tag{4}$$

where $\tilde{P} \in \mathbb{R}^{q\times s\times s\times (1\times t)}$ and $\tilde{Z} \in \mathbb{R}^{q\times 1\times 1\times (b'\times t)}$ represent the multiscale spatial and spectral attention weight vectors, respectively. Finally, we multiply the multiscale feature map $F \in \mathbb{R}^{q\times s\times s\times (b'\times t)}$ by the multiscale spatial and spectral attention weight vectors in turn to generate the weighted multiscale feature map

$$O = \text{Multi}\left(\tilde{Z}, \text{Multi}\left(F, \tilde{P}\right)\right) \tag{5}$$

where $O \in \mathbb{R}^{s\times s\times B}$ $(B = b' \times t \times q)$ represents the weighted multiscale feature map, and Multi represents the element-wise multiplication.

C. Dense Feature Pyramid Network

DFPN is based on dense network and feature pyramid network (FPN), where dense network is an effective structure to solve the problem of information loss and computational burden, and the FPN is widely used to combine both deep and shallow features of DNNs. In fact, a similar structure of DFPN has been applied in the field of remote sensing to improve the accuracy of small object detection [65]. However, we find that the DFPN is not only effective to detect small objects, but also sensitive to different classes of edge pixels in complex backgrounds. There are a large number of edge pixels between known classes and novel class or among different known classes in HSIs, which are often misclassified and affect the final CA. At the same time, DFPN can effectively retain the shallow feature information of the input HSI, while extracting enough feature information for learning. On the one hand, dense network reuses shallow feature information in the feature extraction process. On the other hand, the structure of FPN introduces shallow feature information in the feature reconstruction process. Therefore, DFPN will alleviate the situation that the novel class tends to be classified as the known class with limited network layers and training epochs. In addition, the DFPN also has advantages in multiscale feature fusion, which is easier to adapt to the multiscale feature map weighted by 3D-MAM.

As shown in Fig. 6, the convolution block is the main part of the encoder, which aims to further extract deep semantic feature. The deconvolution block is the main part of the decoder, which aims to reconstruct shallow semantic feature. Specifically, the convolution blocks are densely concatenated to each other, and the concatenated feature maps are added with the corresponding reconstructed feature maps. Each convolution block is composed of a 2-D convolution layer, a batch normalization layer, and a rectified linear unit (ReLU) activation layer, where padding exists in the convolution process. The deconvolution block is composed of a 2-D deconvolution layer, a batch normalization layer, and a ReLU activation layer, where padding also exists in the deconvolution process. Therefore, the feature maps of different blocks in DFPN have the same size, which facilitates feature concatenation or fusion between different blocks.

The process of generating deep feature maps for the weighted multiscale feature map $O \in \mathbb{R}^{s\times s\times B}$ in the encoder

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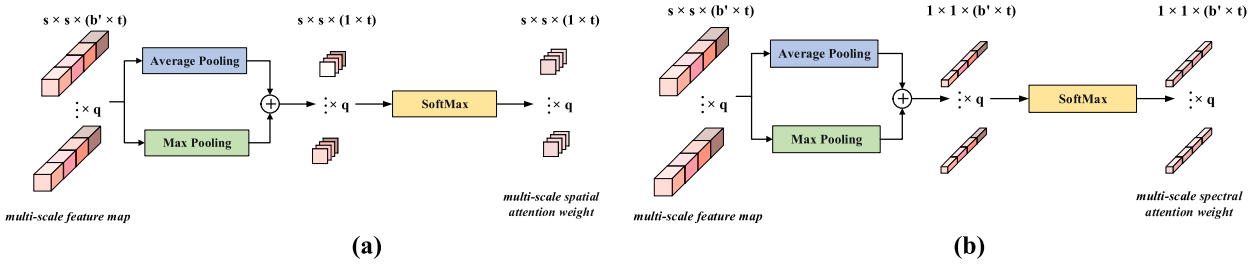


Fig. 5. Schematic of attention blocks with soft attention. (a) Spatial attention block with soft attention. (b) Spectral attention block with soft attention.

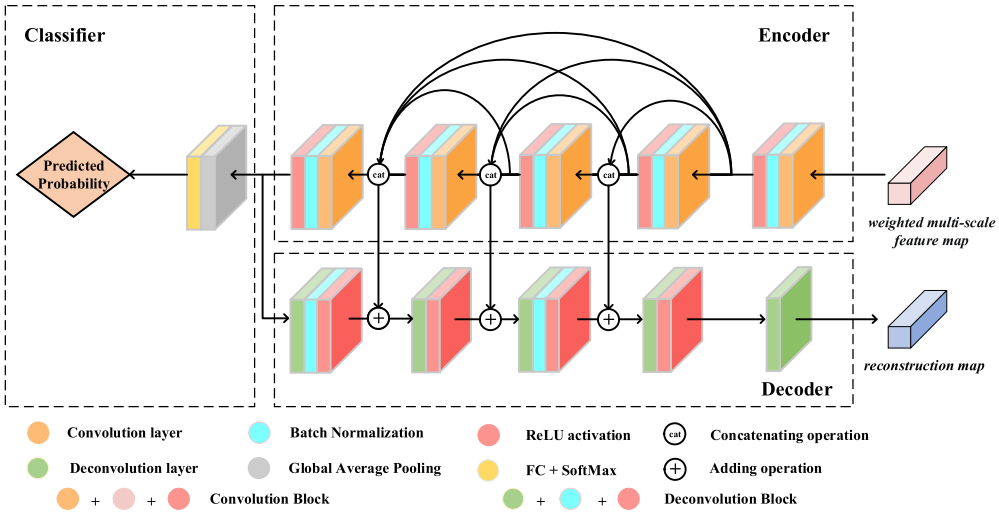


Fig. 6. Schematic of DFPN. DFPN is composed of encoder $\Phi(\cdot)$, decoder $f_r(\cdot)$, and classifier $f_c(\cdot)$.

$\Phi(\cdot)$ containing n convolution blocks can be expressed as

$$\begin{aligned} P_1 &= C_1 = \text{Conv2d}(O) \\ C_j &= \text{Conv2d}\left(P_{j-1}\right), \quad j = 2, \ldots, n \\ P_j &= \text{Concat}\left(C_1, \ldots, C_j\right), \quad j = 2, \ldots, n-1 \end{aligned} \tag{6}$$

where P_1 and $C_j \in \mathbb{R}^{s\times s\times b'}$, $j = 1, \ldots, n$ represents the feature map obtained by the convolution block; $P_j \in \mathbb{R}^{s\times s\times b'}$, $j = 2, \ldots, n-1$ represents the concatenated feature map in DFPN.

The process of reconstructing the shallow feature map for the deep feature map $C_n \in \mathbb{R}^{s\times s\times b'}$ in the decoder $f_r(\cdot)$ containing n deconvolution blocks can be expressed as follows:

$$\begin{aligned} R_1 &= \text{Deconv2d}(C_n) \\ G_j &= \text{Add}\left(R_j, P_{n-j}\right), \quad j = 1, \ldots, n-2 \\ R_j &= \begin{cases} \text{Deconv2d}\left(G_{j-1}\right), & \text{if } j = 2, \ldots, n-1 \\ \text{Deconv2d}(R_{n-1}), & \text{if } j = n \end{cases} \end{aligned} \tag{7}$$

where $R_j \in \mathbb{R}^{s\times s\times b'}$, $j = 1, \ldots, n$ represents the feature map obtained by the deconvolution block; $G_j \in \mathbb{R}^{s\times s\times b'}$, $j = 1, \ldots, n-2$ represents the feature map after feature fusion. The last deconvolution block is only composed of the deconvolution layer without batch normalization layer and ReLU activation layer.

The classifier consists of a global average pooling (GAP) layer and the fully connected (FC) layer with the Softmax activation function. We input the deep feature map $C_n \in \mathbb{R}^{s\times s\times b'}$, and obtain the feature vector ϕ through GAP. Finally, the classifier outputs the predicted probability $p_{i,k}$ that the ith sample belongs to the kth class

$$p_{i,k} = \frac{\exp\left(\phi^T w_k + b_k\right)}{\sum_{k=1}^{K} \exp\left(\phi^T w_m + b_m\right)} \tag{8}$$

where w_k and b_k represent the weight vector and the bias element of the kth neuron in the FC layer, respectively, and K represents the number of the all known category.

D. Loss Function

The proposed framework performs multitask learning based on classification task and reconstruction task. Therefore, we simultaneously jointly use both reconstruction loss and classification loss as the loss function of the model, where the parameters of the model are optimized by minimizing the loss function

$$\Phi(\cdot), f_c(\cdot), f_r(\cdot) = \underset{\Phi(\cdot), f_c(\cdot), f_r(\cdot)}{\arg\min} \left(\lambda_c \ell_c + \lambda_r \ell_r\right) \tag{9}$$

where ℓ_c and ℓ_r represent classification loss and reconstruction loss function, respectively; λ_c and λ_r are the weights controlling the proportion of the classification loss and reconstruction loss, which are generally set to 0.5 [52], [53]. Here, we apply the mean absolute error as the reconstruction loss function

$$\ell_r\left(\tilde{X}, \hat{X}\right) = \left\|\tilde{X} - \hat{X}\right\|_1 \tag{10}$$

where $\tilde{X} = X_{\text{input}}$ represents the input map after DDM and $\hat{X} = R_n$ represents the reconstructed feature map. In addition,

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we apply the cross-entropy loss function as the classification loss function

$$\ell_c\left(\widehat{Y}, Y\right) = -\sum_{k=1}^{K} Y \log\left(\widehat{Y}\right) \tag{11}$$

where K represents the total number of known classes, Y represents the truth label, and $\widehat{Y}$ represents the predicted label that the input sample belongs to the kth class.

E. Adaptive Threshold Module

Although reconstruction-based methods have achieved good results in open-set HSI classification, it takes a lot of time to determine the empirical threshold because of the lack of a principled strategy for adaptive threshold. In fact, adaptive threshold can help to classify the novel class more accurately, which will further improve the performance of the model.

We believe that it is more reliable and effective to classify the novel class from the known class by the uncertainty of predicted probability or reconstruction loss. In fact, the predicted probability of the novel class has been proven to be more uncertain than those of the known classes [60]. However, there are limitations to identifying novel class by relying solely on prediction probability. For example, novel pixels close to known classes tend to be misclassified, which results in an overall decrease in accuracy. This is because the features and predicted probability of these edge pixels belonging to the novel class and known classes are similar and confused. Hence, we define and apply reconstruction loss to further optimize the decision due to a unique conflict of edge pixels. The edge pixels have features closer to the known classes than other pixels of the novel class, while the reconstruction loss of the novel class pixels is always greater than those of the known classes. Therefore, the reconstruction loss of edge pixels belonging to the novel class is more uncertain. In conclusion, a reasonable basis for classifying the novel class is the uncertainty magnitude of predicted probability and reconstruction loss, which can be expressed by entropy.

Finally, there are still three problems to be solved to distinguish novel class: 1) we need to calculate the entropy corresponding to the predicted probability and reconstruction loss; 2) automatic strategy needs to be proposed to determine the threshold of the entropy corresponding to the predicted probability and reconstruction loss; and 3) if the classification results of the same pixel based on the predicted probability and the reconstruction loss are inconsistent, a reasonable criterion needs to be proposed for the final decision.

1) Entropy of Predicted Probability and Reconstruction Loss: We can obtain the predicted probability as $p_{i,k}$ by Softmax, and the uncertainty of the predicted probability of the ith pixel can be represented by entropy

$$\text{Entropy}_{\text{PP}}\left(\tilde{X}_i\right) = -\sum_{k=1}^{K} p_{i,k} \log p_{i,k} \tag{12}$$

where K represents the number of known classes, and $p_{i,k}$ is the probability that the ith pixel belongs to the kth class.

However, the reconstruction loss cannot be directly translated into a probabilistic form $p_{i,k}$, which requires a mathematical definition. Unlike the multiclass classification corresponding to the predicted probability, we take the treatment of reconstruction loss as a binary classification, that is, known classes versus novel class. According to the experience that the greater the reconstruction loss, the more likely the current pixel belongs to the novel class, we count the number of other pixels in the test set whose reconstruction loss is greater than or smaller than the ith pixel, respectively, then the probabilities of known classes $p_{i,\text{known}}$ and novel class $p_{i,\text{unknown}}$ of the ith pixel are defined as follows:

$$\begin{aligned} p_{i,\text{known}} &= \frac{n_{i,\text{pos}}}{n_{i,\text{pos}} + n_{i,\text{neg}}} \\ p_{i,\text{unknown}} &= \frac{n_{i,\text{neg}}}{n_{i,\text{pos}} + n_{i,\text{neg}}} \end{aligned} \tag{13}$$

where $n_{i,\text{pos}}$ represents the number of other pixels whose reconstruction loss is greater than that of the ith pixel, and $n_{i,\text{neg}}$ represents the number of other pixels whose reconstruction loss is smaller than that of the ith pixel. Therefore, the uncertainty of the reconstruction loss at the ith pixel can be expressed by entropy

$$\begin{aligned} \text{Entropy}_{\text{RL}}\left(\tilde{X}_i\right) = &-p_{i,\text{known}} \log p_{i,\text{known}} \\ &- p_{i,\text{unknown}} \log p_{i,\text{unknown}}. \end{aligned} \tag{14}$$

2) Adaptive Threshold Strategies Based on the Uncertainty: The entropy distribution histogram will contain at least two modes (peaks), which provides us with a simple idea to determine the adaptive threshold, that is, the local minimum value (valley position) of the entropy distribution histogram is selected as the threshold θ. Considering further, for the presence of more than two modes (peaks), we choose the maximum of the local minimum in the histogram of the entropy distribution (maximum valley position) as the threshold θ. The reason for choosing the maximum valley as the threshold is that misclassifying pixels of the known classes into the novel class often has a worse impact on the accuracy of the HSI classification, which requires us to increase the threshold for the classification of novel class. Therefore, we automatically determine the boundary threshold θ_{pre} and θ_{res} by finding the valley or maximum valley position on the entropy distribution histogram of predicted probability and reconstruction loss, respectively.

For the uncertainty of the predicted probability, the pixel whose entropy of the predicted probability is greater than the threshold θ_{pre} belongs to the novel class. However, due to the long-tailed distribution characteristics of the reconstruction loss, the above strategy does not work well for the reconstruction loss in our experiments. Inspired by [53], we decided to calculate the tail data size τ and the threshold σ of cumulative distribution function in the generalized Pareto distribution based on the threshold θ_{res}

$$\begin{aligned} \tau &= P \times S \times R \times K \\ \sigma &= \frac{\sum_{i=1}^{T} \text{Entropy}_{\text{RL}}\left(\tilde{X}_i\right)}{\sum_{i=1}^{N} \text{Entropy}_{\text{RL}}\left(\tilde{X}_i\right)} \end{aligned} \tag{15}$$

where P is the number of training samples per class, S is the number of data augmentations types, R represents the fraction

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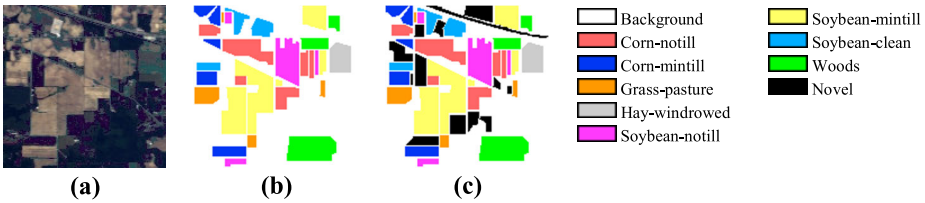


Fig. 7. IP dataset. (a) False-color image. (b) Closed-set ground truth. (c) Open-set ground truth.

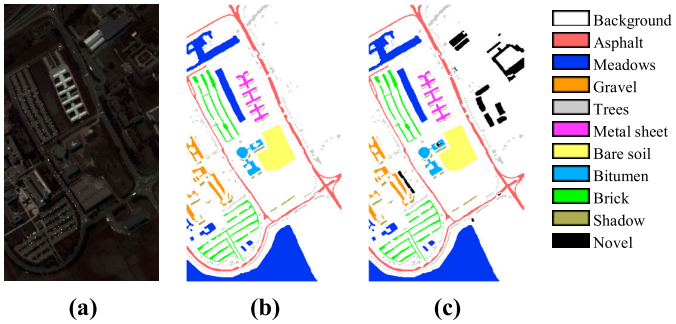


Fig. 8. UP dataset. (a) False-color image. (b) Closed-set ground truth. (c) Open-set ground truth.

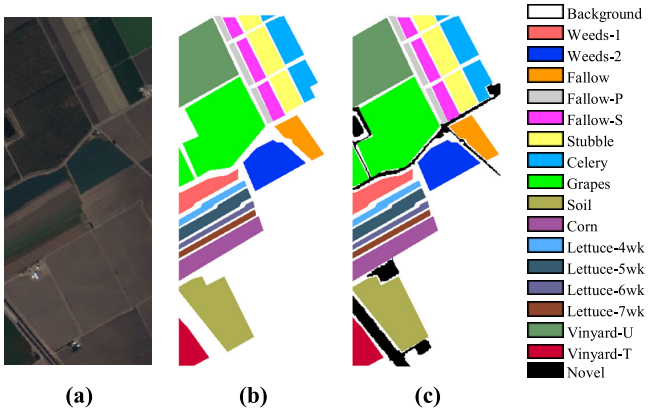


Fig. 9. SA dataset. (a) False-color image. (b) Closed-set ground truth. (c) Open-set ground truth.

of pixels whose entropy is smaller than threshold θ_{res}, K is the total number of all classes, T represents the number of the pixels whose uncertainty is smaller than threshold θ_{res}, and N is the number of all pixels.

In fact, once the adaptive threshold for computing the above two hyperparameters τ and σ is determined, we can classify the novel class by modeling the tail data through extreme value theory (EVT), which is an indirect method that focuses on tail data. Here is a brief explanation of how to use EVT to determine novel class. EVT shows that the distribution of tail data belongs to the Weibull distribution, which means that we can use the Weibull distribution to model the long tail data of reconstruction loss, thus treating outliers that deviate from the distribution as novel class [54], [61]. Finally, the cumulative distribution function $W_{\xi,\mu}(\ell_r)$ based on the Weibull distribution of the reconstructed loss for tail data is as follows:

$$W_{\xi,\mu}(\ell_r) = \begin{cases} 1 - e^{-\frac{\ell_r}{\mu}}, & \xi = 0 \\ 1 - \left(1 + \dfrac{\xi \cdot \ell_r}{\mu}\right)^{-\frac{1}{\xi}}, & \xi \neq 0 \end{cases} \tag{16}$$

where ℓ_r represents the reconstruction loss of $(\tilde{X}, \hat{X})$; ξ and μ are parameters about the tail data [66]. If $W_{\xi,\mu}(\ell_r) < \sigma$, then $\tilde{X}$ belongs to the novel class. Otherwise, $\tilde{X}$ belongs to one of the known classes with the highest predicted probability.

All in all, the above two adaptive threshold strategies classify the novel class from different data processing objects (reconstruction loss and predicted probability) and different data distribution ranges (tail data and global data), respectively. Specifically, we analyze the entropy of the predicted probability from a global perspective and the entropy of the reconstruction loss from a local perspective. Our experimental results show that any single automatic threshold strategy has some limitations, such as unfriendly performance or poor robustness. Besides, one problem may arise if the classification result of the same pixel based on the entropy of reconstruction loss is different from that of predicted probability. Therefore, a reasonable criterion is stipulated that only pixels that are considered known by both strategies are finally classified as known class.

IV. Experiments

A. Experimental Datasets

1) Indian Pines: As shown in Fig. 7, this dataset was collected by the airborne visible / infrared imaging spectrometer (AVIRIS) spectrometer in India, with an area size of 145 × 145 pixels and a total of 200 spectral bands. The false-color image of the dataset is shown in Fig. 7(a). The closed-set dataset of Indian Pines (IP) contains a total of eight known land cover types (since some classes of 16 land cover types have samples fewer than ten, [17] discarded these tail classes, leaving eight known land cover types), among which the number of labeled samples is 8503, and the closed-set ground truth is shown in Fig. 7(b). The IP dataset also contains some samples that should not be classified as any known class with very different spectral profiles, such as roads located in the upper side of the image, which we label as novel class. Therefore, the open-set dataset of IP contains a total of nine classes with a total of 10 510 samples, and the open-set ground truth of the IP dataset manually labeled with novel class is shown in Fig. 7(c).

2) University of Pavia: As shown in Fig. 8, this dataset was acquired by the reflective optics system imaging spectrometer (ROSIS) over the University of Pavia (UP), with an area size of 610 × 340 pixels, a spatial resolution of 1.3 m per pixel, and a total of 103 spectral bands. The false-color image of the dataset is shown in Fig. 8(a). The closed-set dataset of

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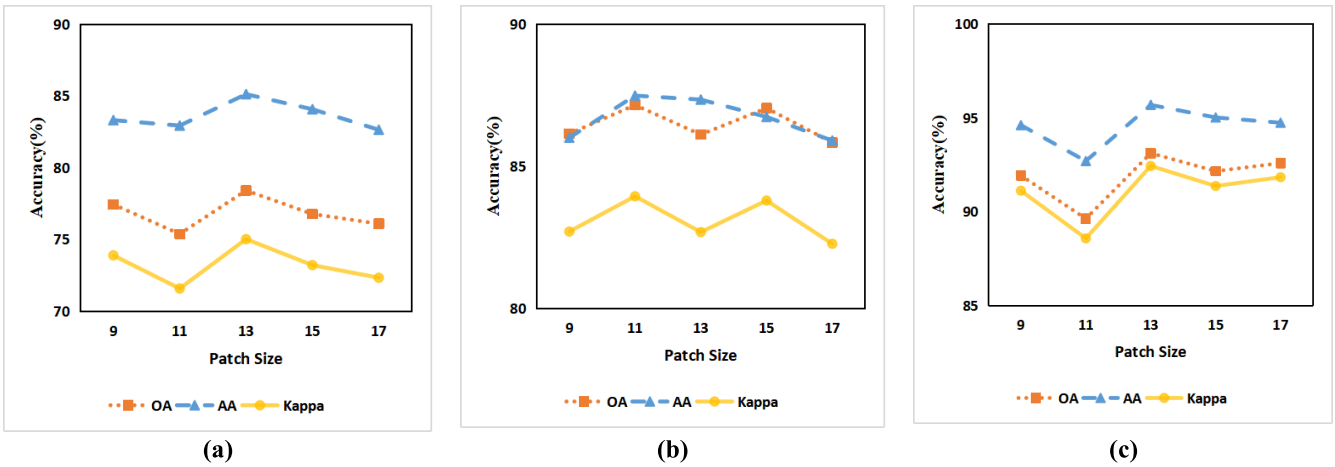


Fig. 10. Results of experiment for hyperparameters. (a) IP dataset. (b) UP dataset. (c) SA dataset.

UP includes nine types of known ground objects in total, among which the number of labeled samples is 42 776, and its closed-set ground truth is shown in Fig. 8(b). There are also some samples in the dataset that should not be classified as any known class with very different spectral profiles, such as some unlabeled buildings, vehicles, and swimming pools, which are manually labeled as novel class. The open-set dataset of UP contains a total of ten classes with a total of 47 939 samples, and the open-set ground truth of UP with novel class manually labeled is shown in Fig. 8(c).

3) Salinas: As shown in Fig. 9, this dataset was collected by the AVIRIS spectrometer in Salinas (SA) Valley, with an area size of 512 × 217 pixels, a spatial resolution of 3.7 m per pixel, and a total of 204 spectral bands. The false-color image is shown in Fig. 9(a). The closed-set dataset of SA contains a total of 16 known land cover types, among which the number of labeled samples is 54 129, and its closed-set ground truth is shown in Fig. 9(b). There are also some samples in the SA dataset that should not be classified as any known class and have very different spectral profiles, such as buildings, roads, and pools, which are manually labeled as novel class. The open-set dataset of SA contains a total of 17 classes with 59 742 samples. The open-set ground truth of the SA with manually labeled novel class is shown in Fig. 9(c).

B. Experimental Settings

The programs of the experiments are implemented in Tensorflow and Keras environment with Python and all results in this article are the average of five experiments on a computer equipped with an NVDIA TITAN RTX.

To verify the performance of the proposed method for few-shot classification, we use only 10 and 20 labeled samples per class as the training data, respectively. The rest of the samples are taken as the testing data. In detail, we use the Adadelta optimizer in the training process. We set the number of spectral bands b' in DDM as 64. The number of convolution kernels q in the four branches of 3D-MAM is eight, and the dimensions k_i are $\{3, 5, 7, 9\}$, respectively, where $i = 1, 2, 3, 4$. The number n of both convolution modules and deconvolution modules in DFPN is set to 5. We choose six different types for data augmentation, where the patches are augmented horizontally, vertically, vortically (both forward and reverse direction), diagonally and unchangeably. Therefore, S is equal to 6. Moreover, the learning rate is set to 1, 0.1, and 0.01 with a decreasing trend, and the model is trained at each learning rate, mainly containing two epochs in the PTS and 200 epochs in fine-tuning stage. The early stopping mechanism is adopted to accelerate the training. If the loss of the model does not decrease for 50 epochs, the training will proceed to the next stage immediately.

Finally, we choose class-specified CA, the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient as evaluation metrics, which are calculated both on the novel and known classes for open-set HSI classification.

C. Experiment for Hyperparameters

The spatial patch size $s \times s$ of input patch X_{input} will have an impact on the performance of the model, which is determined by the characteristics of the dataset itself. As shown in Fig. 10, we test the performance of the model with different spatial patch sizes on the three datasets, where $s \times s$ are set to $\{9, 11, 13, 17\}$, respectively. Since there are branches of convolution size $k_i = 9$ in 3D-MAM, we do not consider the case of spatial patch size smaller than 9. We find that the performance of the model fluctuates with the increase of spatial patch size. Among them, the IP and SA datasets perform best when the spatial patch size is 13, and the PU dataset performs best when the spatial patch size is 11. Therefore, we set the patch size of the three datasets as above to exert the best effect of the model.

D. Experiment for Comparison

To verify the performance of the proposed method SSMTL in few-shot open-set HSI classification, we conducted some experiments with several influential and relevant methods for comparison. On the one hand, it is necessary to compare SSMTL with state-of-the-art (SOTA) OSC methods, which are MDL4OW [52] and SSLR [53]. On the other hand, we compared the proposed method with the conventional CSC

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TABLE I
RESULTS OF EXPERIMENT FOR COMPARISON ON IP WITH 20 AND 10 LABELED SAMPLES PER CLASS

Class	NUM = 20						NUM = 10					
	DFSL	3DCNN	DCCNN	SSLR	MLD4OW	Ours	DFSL	3DCNN	DCCNN	SSLR	MLD4OW	Ours
1	57.39 ±9.46	56.3 ±9.38	62.95 ±11.65	79.10 ±2.70	85.03 ±4.22	**87.9 ±4.21**	55.80 ±8.56	41.96 ±14.76	39.84 ±15.34	61.64 ±15.63	65.84 ±11.52	**66.02 ±10.81**
2	44.70 ±7.50	57.69 ±8.45	56.09 ±23.41	82.26 ±10.94	87.98 ±4.86	**94.65 ±4.12**	44.15 ±12.38	46.51 ±13.1	38.12 ±21.95	74.92 ±8.69	68.19 ±13.48	**87.35 ±10.16**
3	42.48 ±15.01	93.58 ±4.96	58.56 ±25.62	85.47 ±11.63	89.11 ±10.52	**95.49 ±4.26**	29.62 ±2.78	**89.81 ±7.68**	43.82 ±28.58	81.70 ±1.01	81.86 ±4.69	85.8 ±4.16
4	88.81 ±0.77	99.79 ±0.26	86.67 ±7.82	96.19 ±5.51	**99.79 ±0.30**	96.23 ±4.62	89.94 ±1.94	97.53 ±4.95	75.75 ±23.64	**97.99 ±2.21**	98.54 ±1.58	70.92 ±40.41
5	61.42 ±3.25	72.59 ±5.68	58.24 ±16.36	84.22 ±3.84	87.49 ±3.03	**90.74 ±2.07**	52.10 ±7.33	61.67 ±16.2	52.44 ±25.3	77.69 ±4.32	78.07 ±4.19	**85 ±6.54**
6	77.97 ±3.44	62.91 ±12.71	65.88 ±8.14	77.52 ±2.90	74.89 ±5.90	**87.7 ±6.16**	**75.87 ±8.98**	47.42 ±17.34	54.39 ±10.27	62.98 ±6.06	55.6 ±8.75	75.59 ±4.85
7	30.95 ±7.18	66.21 ±5.49	65.74 ±17.48	74.54 ±5.77	84.25 ±2.48	**93.29 ±7.78**	33.15 ±9.70	39.56 ±6.16	48.31 ±20.28	63.17 ±8.21	58.75 ±18.60	**76.66 ±9.82**
8	84.82 ±5.41	95.73 ±2.83	78.64 ±14.92	92.87 ±3.89	93.38 ±3.97	**96.66 ±3.19**	87.66 ±7.81	89.03 ±11.64	72.54 ±20.55	**92.00 ±5.97**	91.31 ±4.07	88.11 ±16.57
Novel	0.00 ±0.00	2.73 ±1.85	4.41 ±2.42	19.53 ±7.10	13.33 ±7.40	**23.38 ±7.86**	0.00 ±0.00	1.83 ±0.57	2.34 ±2.03	18.22 ±9.28	17.96 ±9.66	**21.41 ±14.48**
OA	57.80 ±1.75	58.23 ±2.98	59.14 ±7.1	70.55 ±2.35	71.25 ±0.58	**78.41 ±2.25**	54.79 ±1.53	48.01 ±1.7	43.17 ±7.58	62.51 ±3.03	60.51 ±5.60	**67.57 ±2.39**
AA	54.28 ±1.77	67.5 ±2.2	59.69 ±9.57	76.85 ±2.88	79.47 ±1.16	**85.12 ±0.99**	52.03 ±0.81	57.26 ±4.26	47.51 ±9.87	70.03 ±2.25	68.46 ±5.24	**72.98 ±6.23**
Kappa	52.47 ±1.81	51.88 ±3.58	52.03 ±7.49	66.05 ±2.8	67.07 ±0.64	**71.58 ±1.75**	49.16 ±1.50	40.76 ±2.53	35.18 ±8.93	57.05 ±3.19	55.04 ±6.14	**62.28 ±3.17**

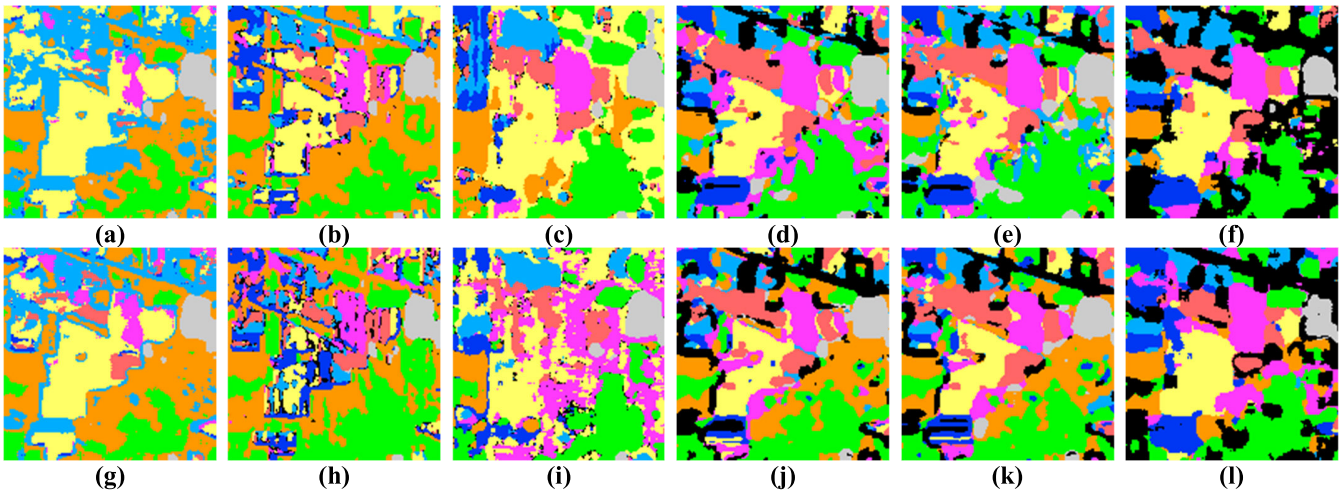


Fig. 11. Classification maps of experiment for comparison on IP. (a) DFSL (NUM = 20). (b) 3DCNN (NUM = 20). (c) DCCNN (NUM = 20). (d) SSLR (NUM = 20). (e) MLD4OW (NUM = 20). (f) Proposed SSMTL (NUM = 20). (g) DFSL (NUM = 10). (h) 3DCNN (NUM = 10). (i) DCCNN (NUM = 10). (j) SSLR (NUM = 10). (k) MLD4OW (NUM = 10). (l) Proposed SSMTL (NUM = 10).

methods to verify the superiority of OSC methods when facing the novel class. In fact, these methods proposed for CSC can also deal with OSC by setting confidence thresholding [67]. Hence, we choose two classical CSC methods, including 3DCNN [18], and deep contextual CNN (DCCNN) [17], where the predicted probability is obtained by FC layer and the confidence thresholding is set to 0.5. This means that the pixels with the maximum predicted probability smaller than 0.5 belong to the novel class.

Besides, we chose deep few-shot learning (DFSL) [57] to verify the performance of the proposed method for few-shot HSI classification, where DFSL is a SOTA few-shot HSI classification method. It should be noted that DFSL cannot predict a novel class because of the closed property of SVM, which is only a CSC experiment in this part. All the above methods are implemented with 10 and 20 labeled samples per class, respectively. Finally, the results of the comparison experiment are shown in Tables I–III and Figs. 11–13, where NUM represents the number of labeled samples used in the experiments.

1) Quantitative Analysis: For the IP dataset, as shown in Table I, we report the class accuracy and three metrics of different methods with 10 and 20 labeled samples, respectively. We find that the proposed method SSMTL performs best. On the one hand, compared with classic CSC methods 3DCNN, DCCNN, and SOTA few-shot HSI classification method DFSL, OA, AA, and Kappa of SSMTL are much larger and the accuracy of the novel class is also improved. On the other hand, even compared with SOTA open-set HSI classification methods SSLR and MDL4OW, the OA of

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TABLE II
RESULTS OF EXPERIMENT FOR COMPARISON ON UP WITH 20 AND 10 LABELED SAMPLES PER CLASS

Class	NUM = 20						NUM = 10					
	DFSL	3DCNN	DCCNN	SSLR	MLD4OW	Ours	DFSL	3DCNN	DCCNN	SSLR	MLD4OW	Ours
1	86.14 ±4.61	65.4 ±9.34	62.71 ±15.09	85.05 ±7.34	**91.34** **±2.97**	87.45 ±6.06	83.14 ±3.64	47.39 ±12.41	60.98 ±17.96	80.99 ±5.34	**83.56** **±5.89**	83 ±2.49
2	87.26 ±5.09	51.09 ±13.62	73.56 ±14.34	**98.17** **±0.88**	96.94 ±1.75	94.92 ±4.49	83.94 ±4.94	54.33 ±20.17	71.03 ±13.19	**89.84** **±6.63**	87.59 ±9.77	86.42 ±9.98
3	79.19 ±5.09	72.06 ±14.09	64.67 ±15.57	86.33 ±6.09	87.28 ±3.98	**90.69** **±3.98**	58.06 ±4.04	71.68 ±10.22	48.38 ±23.38	82.87 ±9.61	79.59 ±13.09	**84.4** **±11.45**
4	93.41 ±3.27	94.92 ±3.41	77.12 ±14.49	**96.04** **±0.94**	93.61 ±2.31	93.69 ±3.47	81.96 ±22.09	94.01 ±4.75	73.95 ±9.24	**93.75** **±2.82**	91.87 ±1.73	90.7 ±4.13
5	83.54 ±14.16	**99.9** **±0.14**	76.33 ±23.21	90.35 ±8.66	87.85 ±3.07	95.61 ±5.02	85.03 ±10.05	**99.87** **±0.2**	75.3 ±18.84	75.08 ±15.20	77.77 ±11.76	81.93 ±34.41
6	56.80 ±5.72	69.76 ±10.41	59.15 ±21.47	93.90 ±4.8	92.20 ±6.18	**97.21** **±2.31**	43.67 ±6.09	58.79 ±22.88	61.45 ±15.49	85.07 ±11.13	87.11 ±7.59	**93.86** **±4.35**
7	57.64 ±10.22	85.2 ±5.25	72.96 ±15.19	97.10 ±1.38	97.86 ±0.13	**98.83** **±1.12**	53.71 ±16.32	82.15 ±5.91	71.36 ±16.7	95.08 ±2.52	96.34 ±2.83	**98.89** **±0.78**
8	69.95 ±6.02	56.07 ±20.01	62.39 ±22.38	83.07 ±6.44	**85.22** **±7.84**	72.37 ±9.03	69.36 ±5.25	50.46 ±14.58	55.66 ±17.86	73.39 ±12.46	**80.14** **±12.67**	65.29 ±4.95
9	73.08 ±21.43	**99.03** **±0.71**	79.3 ±13.4	97.40 ±2.78	95.35 ±1.86	96.94 ±2.92	84.18 ±19.36	95.18 ±4.38	68.97 ±30.29	92.84 ±3.73	93.64 ±2.14	**94.83** **±5.04**
Novel	0.00 ±0.00	1.96 ±1.16	1.56 ±1.39	20.40 ±3.04	18.18 ±3.92	**47.08** **±27.11**	0.00 ±0.00	4.56 ±7.78	2.86 ±2.53	22.14 ±4.2	21.63 ±3.15	**48.94** **±18.34**
OA	69.34 ±1.00	57.1 ±5.05	65.85 ±8.55	85.46 ±1.56	85.39 ±0.73	**87.15** **±1.89**	62.5 ±2.77	54.33 ±6.51	60.55 ±7.24	79.30 ±3.29	79.32 ±4.21	**81.64** **±3.33**
AA	68.7 ±1.81	69.54 ±2.09	62.97 ±11.19	84.78 ±1.82	84.58 ±0.90	**87.48** **±0.51**	64.31 ±3.29	65.84 ±1.72	58.99 ±11.92	79.10 ±2.16	79.92 ±1.92	**82.83** **±2.82**
Kappa	72.59 ±1.35	48.59 ±5.07	59.11 ±9.62	81.71 ±1.97	81.61 ±0.94	**83.94** **±2.27**	63.47 ±3.54	44.7 ±5.74	53.45 ±7.87	74.12 ±4.08	74.37 ±4.83	**77.33** **±3.8**

SSMTL is increased by 7.86% and 7.16% when NUM = 20, and increased by 5.06% and 7.06% when NUM = 10, respectively. The accuracy of the novel class is also increased by 3.85% and 10.05% when NUM = 20, and increased by 3.19% and 3.45% when NUM = 10, respectively. Besides, the AA and Kappa of SSMTL also are higher than those of SSLR and MDL4OW no matter NUM = 20 or NUM = 10.

For the PU dataset, Table II shows the class accuracy and three metrics of different methods with 20 and 10 labeled samples, respectively. It can be found that the proposed method SSMTL performs best. Compared with classic CSC methods 3DCNN, DCCNN, and SOTA few-shot HSI classification method DFSL, OA, AA, and Kappa of SSMTL are much larger and the accuracy of the novel class is also improved. Furthermore, even compared with SOTA open-set HSI classification methods SSLR and MDL4OW, the OA of SSMTL is increased by 1.69% and 1.76% when NUM = 20, and increased by 2.34% and 2.32% when NUM = 10, respectively. The accuracy of the novel class is also increased by 26.68% and 28.90% when NUM = 20, and increased by 26.8% and 27.31% when NUM = 10, respectively. Besides, the AA and Kappa of SSMTL also are higher than those of SSLR and MDL4OW no matter NUM = 20 or NUM = 10.

For the SA dataset, Table III records the class accuracy and metrics of different methods with 20 and 10 labeled samples, respectively. The proposed method SSMTL also performs best. Compared with classic CSC methods 3DCNN, DCCNN, and SOTA few-shot HSI classification method DFSL, SSMTL obtains much larger OA, AA, and Kappa as well as improved accuracy of the novel class. In addition, compared with SOTA open-set HSI classification methods SSLR and MDL4OW, the OA of SSMTL is increased by 4.79% and 3.33% when NUM = 20, and increased by 3.52% and 3.94% when NUM = 10, respectively. The accuracy of the novel class also increased by 27.79% and 21.42% when NUM = 20, and increased by 19.69% and 18.6% when NUM = 10, respectively. Besides, the AA and Kappa of SSMTL also are higher than those of SSLR and MDL4OW no matter NUM = 20 or NUM = 10.

Based on the above analysis, we can draw the following conclusions. First, the proposed method SSMTL can improve the accuracy of novel class while ensuring the accuracy of known classes. As we can observe, SSMTL achieves the largest CA of novel class in all comparison experiments with different datasets, and the accuracy of most known classes is also superior at the same time. Because DFPN proposed in SSMTL is able to extract the deep semantic feature while effectively utilizing the shallow semantic feature, which helps to balance the relationship between the known classes and novel class. Second, the proposed method SSMTL has more robust properties, which can be reflected in the standard deviation of the most metrics. The robustness of SSMTL is due to two factors. On the one hand, in the PTS, the SRS of DDM makes a contribution to the robustness of the model, where the spectral distribution diversity of data is increased. On the other hand, 3D-MAM with Softmax normalization also benefits the robustness of the model. Third, the proposed method SSMTL has more obvious advantages in the case of fewer labeled samples, which can be seen from the results of methods between 20 and 10 samples for different datasets. Compared with other methods, our proposed method performs better with few labeled samples by combining self-supervised learning and multitask learning organically. In conclusion, the above quantitative analysis proves the superior performance of the proposed method SSMTL.

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TABLE III
RESULTS OF EXPERIMENT FOR COMPARISON ON SA WITH 20 AND 10 LABELED SAMPLES PER CLASS

Class	NUM = 20						NUM = 10					
	DFSL	3DCNN	DCCNN	SSLR	MLD4OW	Ours	DFSL	3DCNN	DCCNN	SSLR	MLD4OW	Ours
1	**100.0**	94.49	87.99	94.86	93.73	99.74	**100.0**	97.57	86.14	92.35	92.31	99.6
	±0.00	±7.36	±13.54	±1.69	±0.85	±0.18	**±0.00**	±1.31	±11.83	±3.11	±3.19	±0.33
2	**99.83**	84.91	91.65	97.67	96.84	99.4	**99.65**	66.46	87.51	95.42	95.77	97.36
	±0.29	±6.78	±7.24	±0.98	±0.73	±0.71	**±0.30**	±38.61	±7.27	±3.08	±2.17	±2.8
3	88.14	27.81	85.82	**99.57**	99.13	97.43	89.00	0.00	68.05	93.20	88.95	**98.17**
	±2.51	±38.56	±10.09	**±0.41**	±1.83	±2.1	±3.47	±0.00	±13.62	±8.99	±10.16	**±3.14**
4	85.49	97.65	77.74	99.14	**99.22**	98.05	73.63	90.83	72.01	**98.81**	98.81	94.59
	±18.08	±3.36	±14.16	±0.60	**±0.37**	±0.71	±21.86	±7.75	±24.57	**±0.56**	±0.43	±8.2
5	**97.09**	74.47	75.87	93.31	92.65	95.17	**94.14**	19.88	66.72	89.96	88.63	92.72
	±2.94	±23.7	±15.82	±4.16	±3.88	±1.19	**±3.88**	±22.78	±17.63	±6.29	±5.07	±1.96
6	93.11	98.88	89.58	**99.19**	98.78	99.18	88.80	99.12	91.64	95.80	95.32	**96.99**
	±6.85	±0.98	±7.74	**±0.95**	±1.04	±0.99	±6.78	±0.64	±10.69	±4.52	±4.68	**±3.85**
7	**98.85**	80.55	84.78	97.67	96.64	94.81	**99.04**	86.42	83.14	94.92	95.13	94.88
	±0.91	±24.92	±12.65	±1.36	±0.48	±2.97	**±0.88**	±17.28	±12.78	±3.19	±2.67	±3.47
8	79.5	50.77	85.46	74.14	81.81	**84.84**	79.8	24.73	78.52	76.1	72.43	**81.27**
	±4.25	±33.27	±6.46	±9.23	±3.53	**±4.3**	±5.02	±23.78	±6.04	±6.41	±11.17	**±5.12**
9	96.13	82.21	93.3	**99.79**	99.71	95.51	94.19	53.56	87.89	**99.41**	98.23	96.64
	±2.98	±29.8	±5.66	**±0.24**	±0.44	±2.7	±6.05	±44.01	±14.36	**±0.79**	±2.24	±3.47
10	69.73	73.33	86.83	93.8	**94.32**	91.58	80.58	41.24	81.6	87.08	84.51	**88.54**
	±12.70	±14.01	±6.13	±4.90	**±2.16**	±3.84	±5.51	±17.87	±12.92	±9.83	±8.99	**±2.26**
11	53.87	52.45	73.14	**98.99**	97.85	98.37	46.57	0.00	67.00	96.93	96.57	**98.26**
	±12.71	±47.3	±23.84	**±0.85**	±1.03	±1.36	±14.17	±0.00	±23.04	±1.86	±1.90	**±1.74**
12	95.16	97.31	91.24	99.84	99.65	**99.86**	92.75	40.52	80.62	**99.81**	99.23	98.22
	±3.29	±4.75	±7.25	±0.17	±0.4	**±0.1**	±3.22	±31.76	±15.71	**±0.32**	±1.02	±3.04
13	**100.0**	99.93	88.83	99.37	99.82	99.8	**99.98**	99.52	87.57	99.41	99.89	99.41
	±0.00	±0.1	±9.69	±0.30	±0.20	±0.27	**±0.04**	±0.79	±7.63	±0.28	±0.08	±0.66
14	89.57	94.52	85.18	**99.36**	98.8	98.92	80.93	90.99	84.49	97.91	**99.44**	97.42
	±6.72	±1.46	±13.02	**±0.47**	±1.08	±1.5	±12.97	±2.69	±11.45	±0.97	**±0.5**	±2.23
15	72.78	62.77	84.08	88.58	85.58	**94.93**	67.62	81.73	83.11	79.83	**84.66**	83.05
	±4.67	±34.83	±5.33	±5.36	±5.68	**±3.18**	±3.47	±10.56	±6.61	±8.50	**±7.64**	±8.13
16	51.1	86.93	84.22	98.89	**99.14**	94.96	69.99	70.17	78.55	97.03	**98.16**	94.08
	±24.51	±5.39	±28.3	±0.74	**±0.36**	±2.83	±34.66	±3.88	±27.49	±2.67	**±0.79**	±3.42
Novel	0.00	13.44	1.61	56.10	62.47	**83.89**	0.00	34.02	0.98	67.52	68.61	**87.21**
	±0.00	±3.19	±0.96	±8.69	±5.56	**±16.06**	±0.00	±2.8	±0.59	±9.31	±9.61	**±8.86**
OA	81.80	67.36	82.93	88.32	89.78	**93.11**	80.94	54.39	75.34	87.19	86.77	**90.71**
	±0.71	±2.76	±2.15	±1.23	±0.44	**±0.76**	±0.72	±4.88	±5.25	±0.69	±1.61	**±0.5**
AA	80.61	74.85	80.43	93.55	93.89	**95.68**	79.8	58.63	75.62	91.85	91.57	**94.02**
	±1.51	±4.3	±6.58	±0.47	±0.21	**±0.48**	±1.20	±3.37	±7.37	±1.34	±1.13	**±0.57**
Kappa	80.06	62.86	81.07	87.18	88.77	**92.43**	79.14	46.51	72.76	85.91	85.47	**89.77**
	±0.77	±2.92	±2.43	±1.33	±0.49	**±0.84**	±0.76	±5.06	±5.81	±0.77	±1.73	**±0.55**

2) Visual Analysis: To further visually compare all methods, we show the classification maps of all experiments in Figs. 11–13, where the results with 20 labeled samples per class are shown in the first line of the maps, and the results with ten labeled samples per class are shown in the second line of the maps.

As we can see, there are obvious differences among comparison methods, where the result maps of the proposed method are closest to the open-set ground truth, as shown in Figs. 11(f), 12(f), and 13(f). In fact, the result maps can further prove the superiority of the proposed method over other comparison methods. First, the CSC method DFSL cannot classify the novel class, as shown in Figs. 11(a), 12(a), and 13(a). If we do not add the confidence thresholding to the other two CSC methods 3DCNN and DCCNN, they will also do the same as DFSL. Second, the improved CSC methods with the confidence thresholding also have huge limits. On the one hand, they misclassify many samples of novel class into known classes, as shown in Fig. 13(b) and (h), which reduces the CA of known classes. On the other hand, they ignore many samples of novel class, as shown in Fig. 13(c) and (i), which leads to the result that the model will be insensitive to the novel class. Third, the OSC methods are able to not only ensure the CA of the known classes, but also effectively distinguish the novel class at the same time, as shown in Fig. 13(d) and (e). However, the result maps of two SOTA OSC methods SSLR and MDL4OW are noisy and not smooth enough, as shown in Fig. 11(d), (e), (j) and (k), compared with the result maps of our proposed method, as shown in Fig. 11(f) and (l).

Moreover, our proposed method has outstanding performance in the CA of novel class, as shown in Figs. 11(f), 12(f), and 13(f), and the correctly classified black regions (samples of novel class) are more abundant. All in all, the above visual analysis further proves the superior performance of the proposed method SSMTL.

E. Experiment for Ablation

To further analyze the role of different modules of the proposed method in SSMTL, we conduct ablation experiments using 20 labeled samples per class in IP, PU, and SA. In addition, to facilitate the observation of the performance of each module on different datasets, the patch size of the three datasets in the ablation experiment is all set to 9. In the

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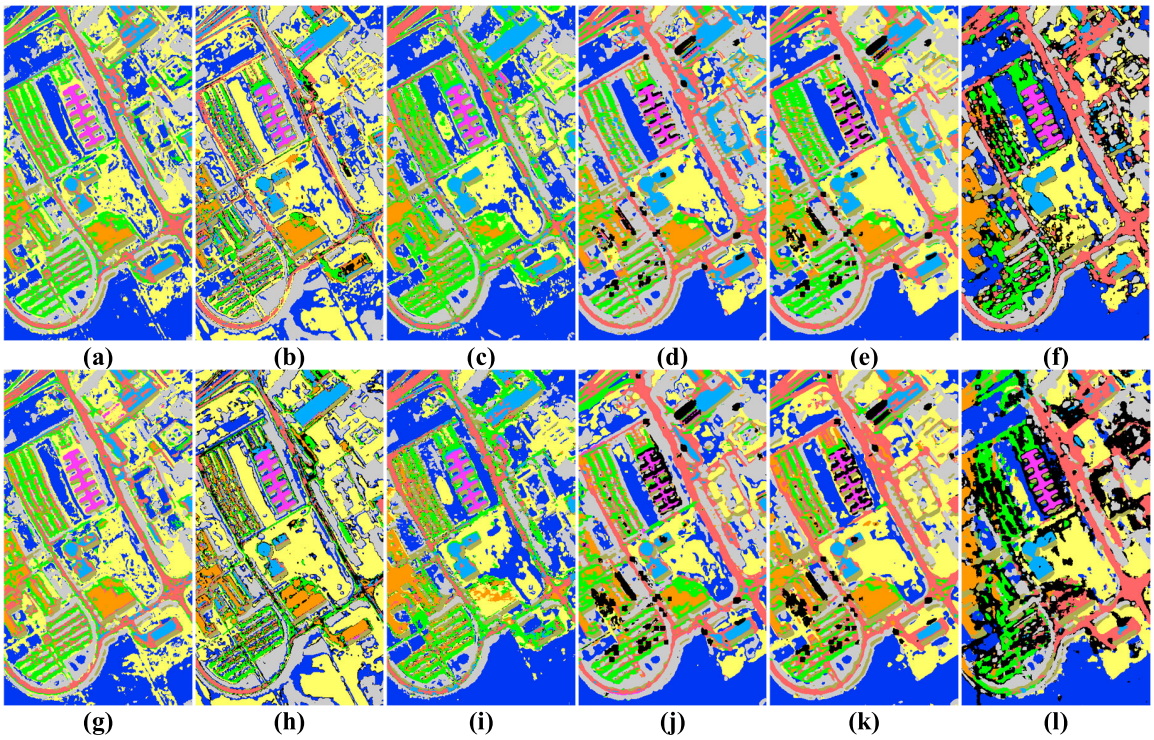


Fig. 12. Classification maps of experiment for comparison on UP. (a) DFSL (NUM = 20). (b) 3DCNN (NUM = 20). (c) DCCNN (NUM = 20). (d) SSLR (NUM = 20). (e) MLD4OW (NUM = 20). (f) Proposed SSMTL (NUM = 20). (g) DFSL (NUM = 10). (h) 3DCNN (NUM = 10). (i) DCCNN (NUM = 10). (j) SSLR (NUM = 10). (k) MLD4OW (NUM = 10). (l) Proposed SSMTL (NUM = 10).

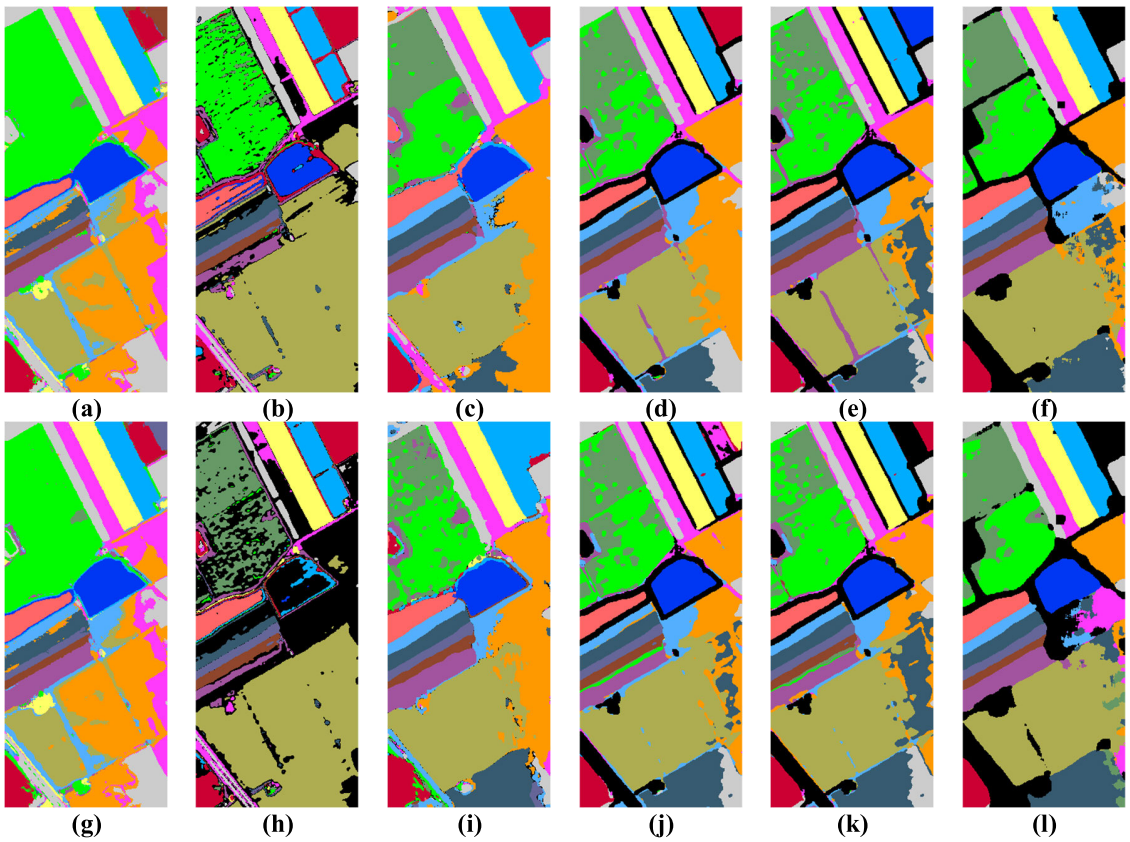


Fig. 13. Classification maps of experiment for comparison on SA. (a) DFSL (NUM = 20). (b) 3DCNN (NUM = 20). (c) DCCNN (NUM = 20). (d) SSLR (NUM = 20). (e) MLD4OW (NUM = 20). (f) Proposed SSMTL (NUM = 20). (g) DFSL (NUM = 10). (h) 3DCNN (NUM = 10). (i) DCCNN (NUM = 10). (j) SSLR (NUM = 10). (k) MLD4OW (NUM = 10). (l) Proposed SSMTL (NUM = 10).

ablation experiments, we explored the effects of PTS, SRS in DDM (SRS-DDM), 3D-MAM, ATM based on predicted probabilities (ATM-PP), and ATM based on reconstruction loss (ATM-RL) in SSMTL separately, which are designed

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TABLE IV
RESULTS OF EXPERIMENT FOR ABLATION ON THREE DATASETS

Modules		Baseline	A	B	C	D	E
PTS		√		√	√	√	√
SRS-DDM		√			√	√	√
3D-MAM		√	√	√		√	√
ATM-PP		√	√	√	√		√
ATM-RL		√	√	√	√	√	
IP	OA	**77.44 ±2.1**	75.38 ±2.14	69.71 ±6.1	73.69 ±2.74	66.34 ±4.43	75.48 ±3.66
	AA	**83.31 ±2.21**	82.59 ±4.38	72.22 ±11.24	78.31 ±5.28	67.93 ±5.82	81.3 ±3.21
	Kappa	**73.89 ±2.38**	71.56 ±2.62	64.33 ±7.59	69.35 ±3.41	65.96 ±9.01	71.44 ±4.29
PU	OA	**85.84 ±0.94**	82.74 ±3.32	80.52 ±6.51	84.53 ±1.87	78.81 ±5.46	83.56 ±2.71
	AA	**86.92 ±1.29**	84.21 ±2.47	79.48 ±10.43	85.98 ±0.64	80.74 ±5.16	79.95 ±6.07
	Kappa	**82.29 ±1.28**	78.55 ±3.9	75.71 ±7.97	80.72 ±2.23	73.86 ±6.64	79.36 ±3.37
SA	OA	**91.92 ±0.61**	89.06 ±3.21	89.39 ±2.14	84.52 ±9.31	91.04 ±2.84	81.3 ±5.85
	AA	**94.60 ±1.06**	92.29 ±3.42	92.45 ±2.9	87.46 ±10.15	93.5 ±3.05	85.64 ±7.26
	Kappa	**91.11 ±0.67**	87.98 ±3.51	88.32 ±2.37	82.96 ±10.24	90.14 ±3.13	79.51 ±6.4

as schemes A–E through different combinations. The details of different schemes are as follows. In schemes A and C, we remove the PTS and 3D-MAM for analyzing the role of corresponding module, respectively, where SRS-DDM itself is included in the PTS. As for scheme B, we replace the SRS with PCA in the PTS to explore the effect of SRS. As for schemes D and E, they are aimed to explore the difference between the single and both adaptive threshold strategies. Besides, we set SSMTL containing all modules as baseline. Finally, the results of comparison experiment are shown in Table IV.

For PTS in SSMTL, compared with scheme A, the results of the baseline show that the performance of the model can get improved with a little training in the PTS; specifically, the OA of three HSI datasets increased by 2.06%, 3.1%, and 2.86%, and the standard deviation of OA decreased by 0.04%, 2.38%, and 2.6%, respectively. Besides, the AA and Kappa of baseline are also higher than those of scheme A, and the overall standard deviation of metrics get decreased as well. In fact, although the results of scheme A do not show dramatic changes, pre-training does achieve improvements on all three datasets with limited training.

For SRS-DDM in SSMTL, compared with scheme B, the results of the baseline show that random sampling can improve the accuracy and robustness of the model. Specifically, the OA of three HSI datasets increased by 7.73%, 5.32%, and 2.53%, and the standard deviation of OA decreased by 4.0%, 5.57%, and 1.53%, respectively. Besides, the AA and Kappa of baseline are also higher than those of scheme B and the overall standard deviation of metrics get decreased as well. In fact, compared with scheme A, the results of scheme B on IP and PU datasets are not satisfactory, which indicates that simple pre-training does not always improve model performance and it is a wise choice to employ SRS for pre-training.

For 3D-MAM in SSMTL, compared with scheme C, the results of the baseline show that the proposed AM module can improve the accuracy and robustness of the model. Specifically, the OA of three HSI datasets increased by 3.75%, 1.31%, and 7.4%, and the standard deviation of OA decreased by 0.64%, 0.93%, and 8.7%, respectively. Besides, the AA and Kappa of baseline are also higher than those of scheme C and the overall standard deviation of metrics get decreased as well. In fact, the AM combined with Softmax normalization can improve the robustness of the model to a certain extent.

For ATM-PP in SSMTL, compared with scheme D, the results of the baseline show that the performance of model can get improved with both adaptive threshold strategies. Specifically, the OA of three HSI datasets increased by 11.1%, 7.03%, and 0.88%, and the standard deviation of OA decreased by 2.33%, 4.52%, and 2.23%, respectively.

For ATM-RL in SSMTL, compared with scheme E, the results of the baseline show that the performance of model can get improved with both adaptive threshold strategies. Specifically, the OA of three HSI datasets increased by 1.96%, 2.28%, and 10.62%, and the standard deviation of OA decreased by 1.56%, 1.77%, and 5.24% respectively. Besides, the AA and Kappa of baseline are also higher than those of schemes D and E and the overall standard deviation of metrics get decreased as well.

Scheme D makes better performance than scheme E on the SA dataset, while it turns to be the opposite on the IP and UP datasets. ATM-RL can further optimize the classification results especially on SA dataset obtaining larger novel area and more edge pixels. On IP and UP datasets with fewer edge pixels, ATM-PP obviously plays a more critical role. In fact, there is no absolute superiority between the two strategies, which perform differently on different datasets.

All in all, the above ablation experiment proves that different modules of the proposed method SSMTL have important roles in the OSC task, where the majority of modules can have an impact on the CA and robustness of the model.

F. Analysis of Backbone Network

To further verify the advantages of the backbone network DFPN of the proposed method, we select some HSI classification methods with different network structures for comparison. At the same time, all experiments are uniformly carried out under the traditional closed-set HSI classification assumption to avoid other interference. Specifically, we add six new comparison methods with different backbone network, including GCN-based methods: GCN [25] and spatial pooling GCN (SPGCN) [27], transformer-based methods: spectral–spatial transformer network (SSTN) [30] and spectral former [31], CNN-based methods: double-branch dual-attention network (DBDA) [21] and 3-D octave convolution with spatial–spectral attention network (3DOC-SSAN) [23].

As shown in Table V, the proposed DFPN performs best on IP and SA and has the ability to compete with other methods on UP. As we can observe, the performances of traditional CNN-based methods are not satisfactory for few-shot HSI classification, such as 3DOC-SSAN, while transformer-based

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TABLE V
RESULTS OF ANALYSIS OF BACKBONE NETWORK ON THREE DATASETS

Methods		IP			UP			SA		
		OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
GCN-based	GCN	60.86 ±1.35	71.03 ±0.64	56.13 ±1.38	65.19 ±3.36	79.04 ±1.18	57.75 ±3.51	86.67 ±0.56	93.03 ±0.18	85.24 ±0.59
	SPGCN	88.08 ±2.55	93.98 ±1.13	86.55 ±2.82	89.19 ±1.9	93.26 ±1.26	86.02 ±2.44	93.17 ±1.58	97.2 ±0.6	92.42 ±1.76
Transformer-based	SSTN	91.01 ±0.86	94.94 ±0.38	89.79 ±0.98	93.55 ±0.72	**94.97 ±0.94**	91.53 ±0.94	94.65 ±0.66	97.6 ±0.18	94.06 ±0.73
	Spectral Former	62.3 ±3.17	75.29 ±2.04	57.89 ±3.19	64.8 ±3.14	75.06 ±1.47	56.37 ±3.19	81.88 ±0.89	89.26 ±1.05	79.97 ±0.95
CNN-based	DBDA	89.19 ±1.13	79.10 ±0.95	87.73 ±1.26	**94.75 ±0.92**	93.17 ±0.95	**93.11 ±1.17**	90.47 ±6.54	95.32 ±2.58	89.48 ±7.12
	3DOC-SSAN	68.92 ±3.41	80.27 ±2	65.02 ±3.79	78.93 ±2.14	77.23 ±2.58	71.88 ±2.88	88.65 ±1.91	94.33 ±0.78	87.39 ±2.13
	Proposed DFPN	**91.32 ±1.43**	**95.38 ±0.75**	**90.17 ±1.61**	91.01 ±1.92	93.33 ±1.35	88.32 ±2.40	**94.89 ±0.65**	**97.86 ±0.25**	**94.32 ±0.72**

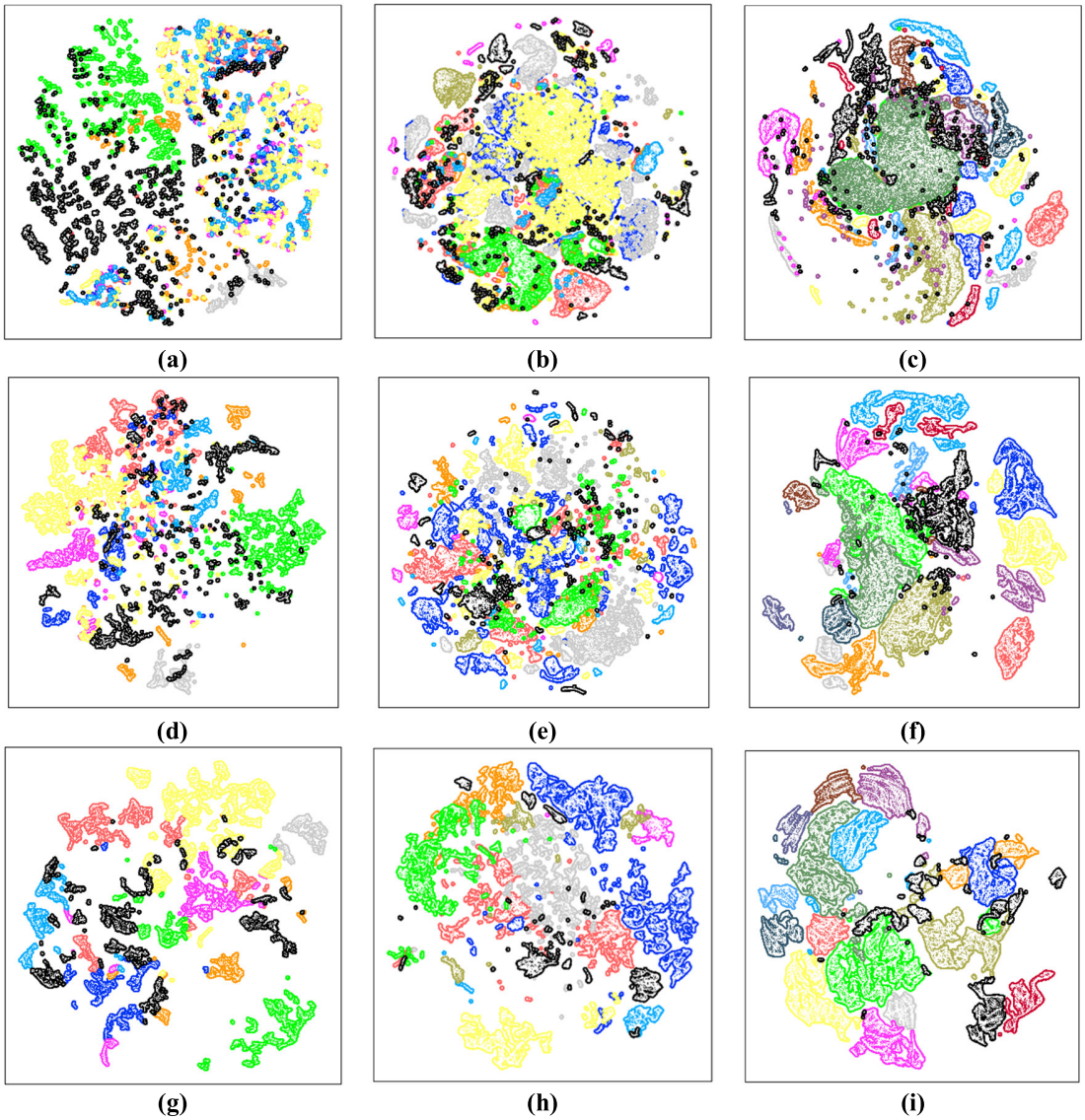


Fig. 14. t-SNE visualization results of feature space. (a) Original feature space on IP. (b) Original feature space on UP. (c) Original feature space on SA. (d) Feature space constructed by SSLR on IP. (e) Feature space constructed by SSLR on UP. (f) Feature space constructed by SSLR on SA. (g) Feature space constructed by SSMTL on IP. (h) Feature space constructed by SSMTL on UP. (i) Feature space constructed by SSMTL on SA.

and GCN-based methods seem to be better, such as SSTN and SPGCN. In fact, DFPN is also a network structure based on CNN. However, even compared with other advanced network structures, DFPN is still on par with them. Hence, there is no doubt that the proposed backbone network has potential and advantages for few-shot HSI classification.

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TABLE VI
COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS

Datasets	Computational Complexity	3DCNN	DCCNN	SSLR	MLD4OW	Ours (Model Inference + adaptive threshold inference)
IP	Number of parameters	92,333	905,736	362,575	494,864	1,702,792
	Inference Time (s)	4.29	7.13	24.88	25.85	7.21+11.56
PU	Number of parameters	65,874	493,961	278,735	383,088	1,702,857
	Inference Time (s)	25.93	28.31	57.77	56.79	61.71+229.04
SA	Number of parameters	152,821	640,272	366,555	499,996	1,703,312
	Inference Time (s)	22.28	21.31	50.66	49.58	36.19+97.12

G. Analysis of Feature Space Visualization

To intuitively show the performance of the SSMTL in the feature space for OSC, as shown in Fig. 14, we adopt the t-distributed stochastic neighbor embedding (t-SNE) dimension-reduction visualization approach to visually show the changes in the feature space [56], where the color settings of different categories in the visualization result are the same as before. As we can observe, the distribution of the original data in the feature space is disorderly, where the samples of different known classes or novel class are interleaved. However, the feature space constructed by SSMTL on the three datasets clearly becomes more regular and fits the purpose of classification. Even compared with the results of SSLR, one of the SOTA OSC methods, our results are still better on the distribution of feature space of the same class or different classes. On the one hand, the samples belonging to the same class show clustering phenomena, and there are clear edges between different classes. Hence, the distances of samples belonging to the same category are narrowed in the feature space constructed by SSMTL, which is beneficial to classifying the known classes. On the other hand, samples belonging to the novel class can be distinguished from known classes to some extent. Hence, the distances of samples belonging to known classes and novel class are expanded in the feature space constructed by SSMTL, which is beneficial to classifying the novel class. Besides, the adaptive threshold makes the feature space of novel class present a multicenter clustering phenomenon, which is beneficial to identifying the novel class as much as possible without affecting the CA of the known classes.

H. Analysis of Computational Complexity

In this section, we further analyze the computational complexity of the proposed SSMTL. Specifically, we compare the number of parameters and inference time of SSMTL with other methods, where DFSL is not included because DFSL cannot predict a novel class and is only carried out as a CSC experiment. It is noted that all methods are performed using the NVIDIA TITAN RTX GPU and the Intel Xeon E5-2630 v4 CPU. As shown in Table VI, the proposed SSMTL achieves the highest CA within a reasonable inference time by using a larger number of parameters, where most of the inference time is consumed by determining the adaptive threshold. SSMTL contains some effective modules, which increase the computational complexity to some extent, but this is acceptable under most computing platforms and application scenarios, considering the significant improvement in CA.

V. CONCLUSION

Although deep learning-based HSI classification methods have made great progress, there are few robust and effective methods for the open-set HSI classification. In addition, the existing open-set HSI classification methods also have limitations in terms of few-shot problem, empirical threshold, and robustness. This article proposes an SSMTL framework for few-shot open-set HSI classification. Specifically, we construct a self-supervised reconstruction task and a classification task with a few labeled samples for few-shot problems and propose two adaptive threshold strategies based on the uncertainty for open-set problems. In addition, we design several functional modules and a backbone network to improve the CA of the novel class and the robustness of the model while ensuring the accuracy of known classes. More importantly, different functional modules and backbone network can be combined organically with an efficient learning paradigm SSMTL to achieve diverse performance gains simultaneously.

To verify the performance of the proposed method SSMTL, we conduct sufficient experiments on three open-set HSI datasets. The experimental results show that the proposed method can effectively classify novel class and ensure the CA of known classes, which has advantages compared with SOTA closed-set and open-set HSI classification methods. However, we note that the model cannot fully recognize information about novel class, and methods based on reconstruction are still relatively unstable. In the future, we will try to design more stable and robust few-shot open-set HSI classification models.

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